

Alpha Diversity

Muluh

Community diversity

Group-wise comparisons

- **Index:** observed
- **Group variable:** meal_group
- **Multiple testing correction:** fdr
- **Statistical test** Wilcoxon

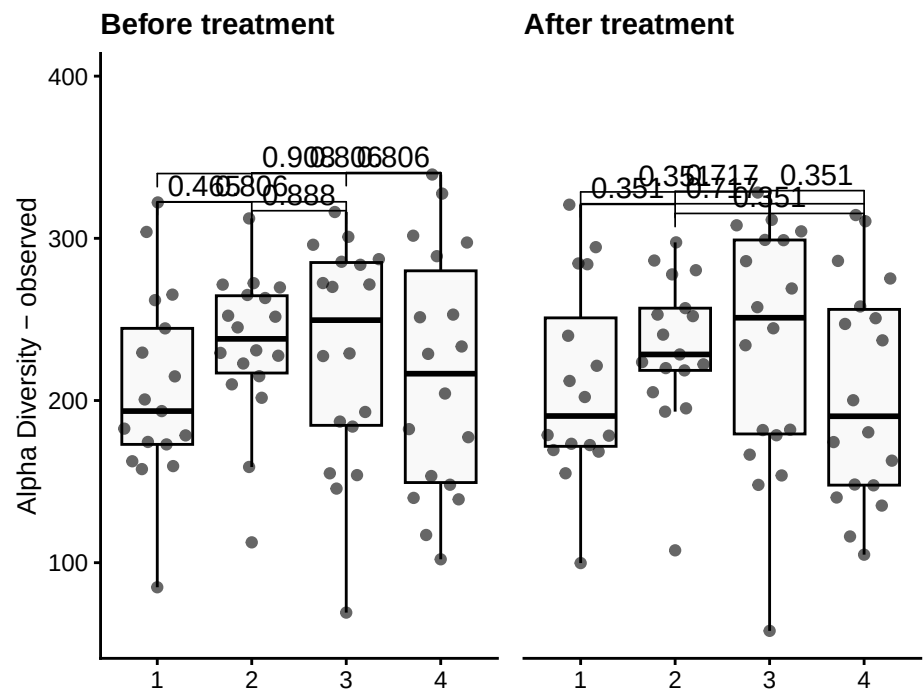


Table of the top findings

Across-group

Before treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 1: Table of alpha diversity comparisons before Treatment

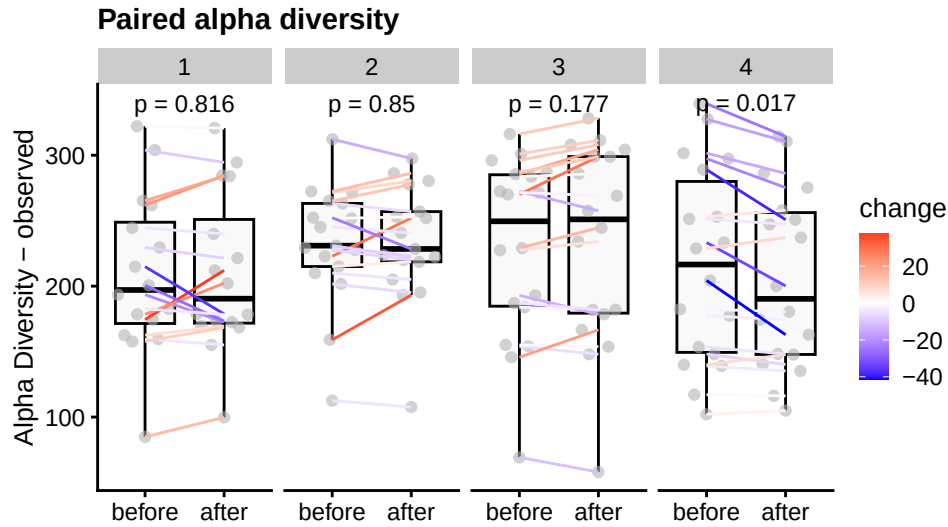
group1	group2	mean1	mean2	log2FC	p.adj
1	2	206.4	234.0	0.18	0.465
1	3	206.4	229.3	0.15	0.806
1	4	206.4	215.8	0.06	0.908
2	3	234.0	229.3	-0.03	0.888
2	4	234.0	215.8	-0.12	0.806
3	4	229.3	215.8	-0.09	0.806

After treatment

- Multiple testing correction: **fdr**
- Statistical test: **Wilcoxon**

Table 2: Table of alpha diversity comparisons after Treatment

group1	group2	mean1	mean2	log2FC	p.adj
1	2	209.7	232.8	0.15	0.351
1	3	209.7	233.9	0.16	0.351
1	4	209.7	205.0	-0.03	0.717
2	3	232.8	233.9	0.01	0.717
2	4	232.8	205.0	-0.18	0.351
3	4	233.9	205.0	-0.19	0.351



Within-group

Paired treatment

- Multiple testing correction: **paired** -none
- Statistical test: **Wilcoxon**

Table 3: Table of alpha diversity paired between timepoint(1 and 2)

meal_group	mean1	mean2	log2FC	p.value
1	206.4	209.7	0.06	0.816
2	234.0	232.8	0.07	0.850
3	229.3	233.9	0.02	0.177
4	215.8	205.0	-0.06	0.017

Alternate analyses

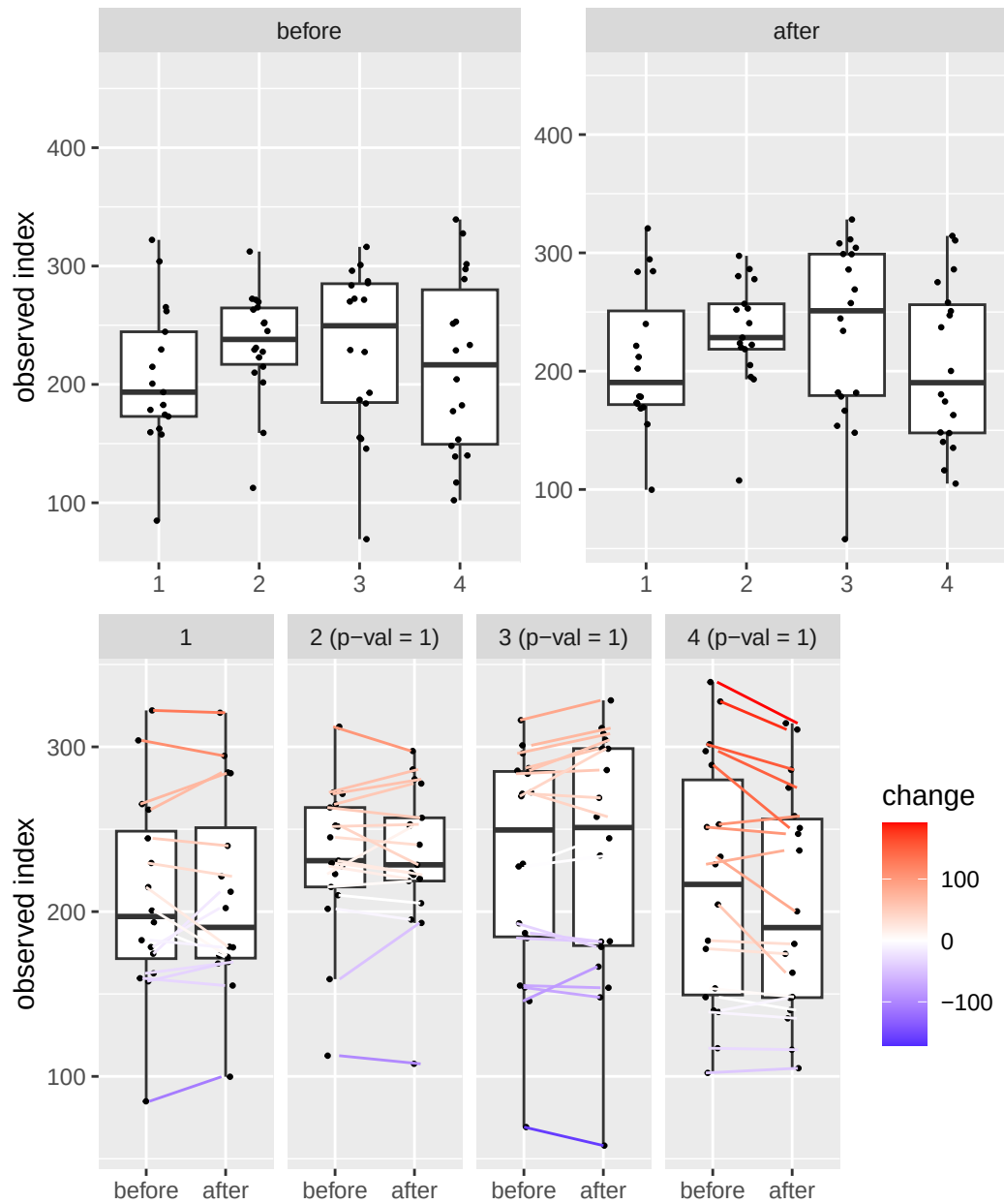
- **Beta (Estimate):** Shows the effect size and direction of a predictor on the outcome. Positive means an increase, negative a decrease.
- **95% CI:** Range where the true beta is likely to fall, reflecting precision.
- **p-Value:** Tests if beta differs from zero; $p < 0.05$ suggests significance.

Characteristic	before			after		
	Beta	95% CI ¹	p-value	Beta	95% CI ¹	p-value
(Intercept)	151	67, 236	<0.001	144	59, 229	0.001
meal_group			0.4			0.3
1	—	—		—	—	
2	32	-11, 75	0.14	28	-16, 72	0.2
3	28	-15, 71	0.2	30	-14, 73	0.2
4	10	-32, 53	0.6	-3.9	-47, 39	0.9
age	1.0	-0.44, 2.5	0.2	1.2	-0.25, 2.7	0.10

¹CI = Confidence Interval

Paired test between timepoints (1 and 2)	Beta	95% CI ¹	p-value
(Intercept)	147	-Inf, Inf	<0.001
meal_group			0.4
1	—	—	
2	28	-Inf, Inf	>0.9
3	28	-Inf, Inf	>0.9
4	2.1	-Inf, Inf	>0.9
age	1.2	0.14, 2.2	0.025

¹CI = Confidence Interval



Data Summary

```
class: TreeSummarizedExperiment
dim: 1700 140
metadata(0):
assays(2): counts relabundance
rownames(1700): SGB4933 SGB4285 ... SGB15018 SGB4921
```

```
rowData names(9): kingdom phylum ... strain clade_name
colnames(140): AE1332 AK1371 ... TS2358 VK2336
colData names(13): sample id ... shannon group
reducedDimNames(0):
mainExpName: NULL
altExpNames(10): PrevalentGenus PrevalentSpecies ... species strain
rowLinks: NULL
rowTree: NULL
colLinks: NULL
colTree: NULL
```